

Aaron Leevord Fernandes

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SUMMARY AI and ML Engineer with experience building distributed backend systems, AI infrastructure, and high-performance computing applications. Experienced developing backend systems, scalable data pipelines, GPU-accelerated machine learning systems, and cloud-native applications using Python, Java, C++, AWS, Docker, and Linux. Strong background in machine learning research, computer vision, RL, ML systems, distributed training, and backend engineering, with expertise spanning production software, AI infrastructure, and large-scale experimentation.

EDUCATION

- Indiana University Bloomington** — M.S. Intelligent Systems Engineering — **GPA: 3.9/4.0** May 2025
- Coursework: Computer Vision, Deep Learning Systems, Reinforcement Learning, NLP, Cloud Computing, Multivariate Statistics
 - Research Areas: Distributed ML Systems, Computational Imaging, Reinforcement Learning, Generative AI
- Goa College of Engineering** — B.S. Electronics and Telecommunications Engineering — **GPA: 3.7/4.0** Sept 2021
- Research Areas: Computer Networking, IoT

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript/TypeScript, SQL

Backend & Systems: Spring Boot, FastAPI, REST APIs, GraphQL, PostgreSQL, MongoDB, Redis, Linux, Docker, Git, CI/CD

Cloud & Distributed Computing: AWS (EC2, S3, Lambda, SageMaker), Kubernetes, Slurm, Spark, Hadoop, MPI

ML Systems: PyTorch, TensorFlow, HuggingFace, ONNX, DeepSpeed, DDP, Triton, Ray, vLLM, MLflow, Weights & Biases

AI Domains: Computer Vision, Diffusion Models, Vision Transformers, LLMs, Reinforcement Learning, RAG

EXPERIENCE

- Graduate Research Assistant — Synthetic Driving Data for Autonomous Driving (Industry Sponsored) — Indiana University** Mar 2026 – Present
- Built distributed GPU inference and benchmarking pipelines for evaluating diffusion-based world models on large-scale autonomous driving datasets.
 - Designed scalable preprocessing, dataset curation, and experiment orchestration workflows supporting multimodal training and evaluation pipelines.
 - Automated benchmarking, profiling, and validation workflows to improve reproducibility and accelerate experimentation across distributed GPU infrastructure.
 - Analyzed system performance, throughput, robustness, and model behavior under distribution shifts to guide production-ready deployment decisions.
- Software Engineer — Tata Consultancy Services (HDFC Bank)** Jan 2022 – Jan 2023
- Developed and maintained Spring Boot microservices supporting **100K+** daily financial transactions in a regulated enterprise banking environment.
 - Designed and maintained RESTful backend services, collaborating across development, QA, and infrastructure teams to deliver reliable production software.
 - Managed production deployments, release validation, CI/CD workflows, and issue resolution across distributed enterprise systems.
 - Improved maintainability and operational stability through backend enhancements, production support, and cross-functional engineering collaboration.
- Machine Learning Engineer (Computer Vision) — Leung Research Group** Apr 2024 – Dec 2025
- Developed GPU-accelerated computer vision pipelines for computational imaging and OCT-to-histology translation using PyTorch.
 - Engineered scalable HPC workflows for optical simulations on Linux clusters using SLURM, improving throughput for large-scale experimentation.
 - Built surrogate deep learning models reducing optical simulation latency from **~3 minutes to sub-second inference** while maintaining **SSIM > 0.85**.
 - Designed reproducible training, evaluation, and data-processing pipelines improving experiment scalability and model development efficiency.
- Machine Learning Engineer (Reinforcement Learning) — Indiana University** Jan 2025 – Jun 2025
- Implemented scalable reinforcement learning infrastructure for goal-conditioned policy optimization using HER and curriculum learning.
 - Built distributed experimentation pipelines using Hydra, MLflow, and Weights & Biases to orchestrate **100+** multi-seed training runs.
 - Automated configuration management, experiment tracking, benchmarking, and hyperparameter optimization to improve reproducibility across RL workloads.

PROJECTS

- ScoutVideo | Python, LangGraph, FastAPI, vLLM, Cosmos-Reason2-8B, Qwen3-8B, Whisper, FalkorDB, PostgreSQL, Graph-RAG, Langfuse, Docker** 2026
- Built a real-time, full-duplex voice agent (mic → STT → LLM → TTS) that watches video and converses; WebAudio AudioWorklet 16 kHz PCM streaming, energy VAD, barge-in, and sentence-streaming TTS.
 - Architected an agentic tool-calling orchestrator in LangGraph/LangChain (durable PostgreSQL checkpointing, stateful multi-turn threads): a Qwen3-8B LLM autonomously invokes specialist tools via OpenAI-compatible function calling in an llm-tools loop, bounded by a deterministic zero-temperature router for reliability.

- Served Cosmos-Reason2-8B (VLM) and Qwen3-8B on vLLM across GPU servers over SSH tunnels; routing visual queries to the vision model removed an LLM hallucination failure mode, streaming answers back verbatim.
- Implemented Graph-RAG over FalkorDB (Cosmos-Embed1 256-dim vector KNN + graph-hop filters in hybrid Cypher) and fused nuScenes telemetry as ground truth for motion.
- Dockerized FalkorDB + PostgreSQL/pgvector behind FastAPI/WebSockets; added Langfuse tracing with media masking for end-to-end observability.

Grounded Theory Analysis Automation Platform — Electron, React, LangChain, Neo4j

Oct 2024 – Dec 2024

- Developed a cross-platform desktop application using Electron and React for qualitative research workflows.
- Built scalable document ingestion, retrieval, and knowledge graph pipelines using Neo4j, LangChain, LlamaIndex, and vector embeddings.
- Designed interactive visualization and querying interfaces enabling researchers to explore large document collections efficiently.
- Reduced qualitative analysis effort from months to days by automating approximately **90%** of the coding workflow.

Diffusion Model Training Infrastructure — PyTorch, DDP, DeepSpeed, SLURM

Sept 2025 – Present

- [Github: Building Diffusion Models](#)
- Built a modular diffusion model training framework from first principles with configurable U-Net architectures, timestep schedulers, objectives, and sampling algorithms for reproducible experimentation.
- Designed reusable ML infrastructure supporting checkpointing, mixed-precision training, experiment configuration, logging, and deterministic evaluation across multiple experiments.
- Implemented distributed multi-GPU training using DDP and DeepSpeed, profiling GPU utilization, communication overhead, memory consumption, and data-loading bottlenecks to optimize training throughput.
- Developed automated benchmarking pipelines measuring convergence, image quality, inference latency, GPU efficiency, and training stability across diffusion model variants.
- Engineered scalable preprocessing and streaming pipelines for a **300+ GB** LAION-Aesthetics image-text dataset supporting high-throughput distributed training.

ComicPaliGemma — Vision Language Model — PyTorch, CUDA, Triton

July 2024 – Present

- Implemented Google's PaliGemma architecture from scratch by integrating SigLIP vision encoder and Gemma decoder while maintaining pretrained checkpoint compatibility.
- Fine-tuned the model on a **5K+** comic-panel dataset using **QLoRA (4-bit)**, reducing peak GPU memory consumption by **65%**.
- Developed optimized autoregressive inference pipelines with Triton/FlashAttention-style attention kernels achieving **5–10×** faster inference compared to standard PyTorch implementations.
- Designed modular training, inference, and evaluation pipelines enabling efficient experimentation with multimodal architectures.

Serverless Distributed Matrix Multiplication — Google Cloud Platform

Aug 2024 – Sept 2024

- Designed a distributed MapReduce framework using Google Cloud Functions, Pub/Sub, and Memorystore for scalable matrix multiplication.
- Implemented asynchronous worker coordination, distributed task execution, and fault-tolerant communication across cloud services.
- Optimized workload partitioning and communication overhead, improving execution efficiency by **25%** for matrices up to **10 GB**.

Model-Based Reinforcement Learning with MPC — PyTorch, MuJoCo

Aug 2024 – Dec 2024

- Implemented a model-based reinforcement learning framework integrating learned dynamics models with Model Predictive Control and Cross-Entropy Method optimization.
- Built reusable planning, simulation, and evaluation components achieving approximately **2×** sample efficiency over SAC baselines on HalfCheetah and Reacher.

Booking Management Platform — Next.js, FastAPI, MySQL, Stripe

June 2024 – Aug 2024

- Developed a full-stack booking platform with secure authentication, role-based access, real-time chat, email notifications, and Stripe payment integration.
- Designed and implemented RESTful backend APIs for bookings, payments, and user management, improving frontend-backend communication and transaction reliability.
- Built and tested **10+ production features** with automated unit testing, reducing production defects by **40%**.
- Optimized database interactions and application workflows to improve responsiveness and user experience.

CERTIFICATIONS

[AWS Certified Developer – Associate](#)

[AWS Certified Cloud Practitioner](#)

OPEN SOURCE

Maintainer of personal machine learning repositories focused on diffusion models, vision-language models, reinforcement learning, and scalable AI systems with reproducible training infrastructure.

Actively develop reusable PyTorch implementations emphasizing modular design, benchmarking, distributed training, and reproducible experimentation.